

Chapter 3

Molecular interactions

A mini review

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Molecular interactions

The fundamental concept of a drug working in an organism lies in a simple question: how a small molecule finds and retains to its protein target. The answer is molecular recognition, which is the ability of two distinct entities: a small chemical compound and a big biomolecule interacting selectively with each other through precise geometric and electronic complementarity. When a drug binds to its target, usually there is no covalent bonding (an exception is an irreversible inhibition explained in Pharmacodynamic Chapter), but relying instead on non-covalent interactions *via* hydrophobic contact, hydrogen bonding, electrostatic interactions and others. Each of them individually is weak, easy to break. The molecular recognition depends not on a single interaction but a sum of many complementary contacts across the ligand-receptor interface (Fig. 1).¹

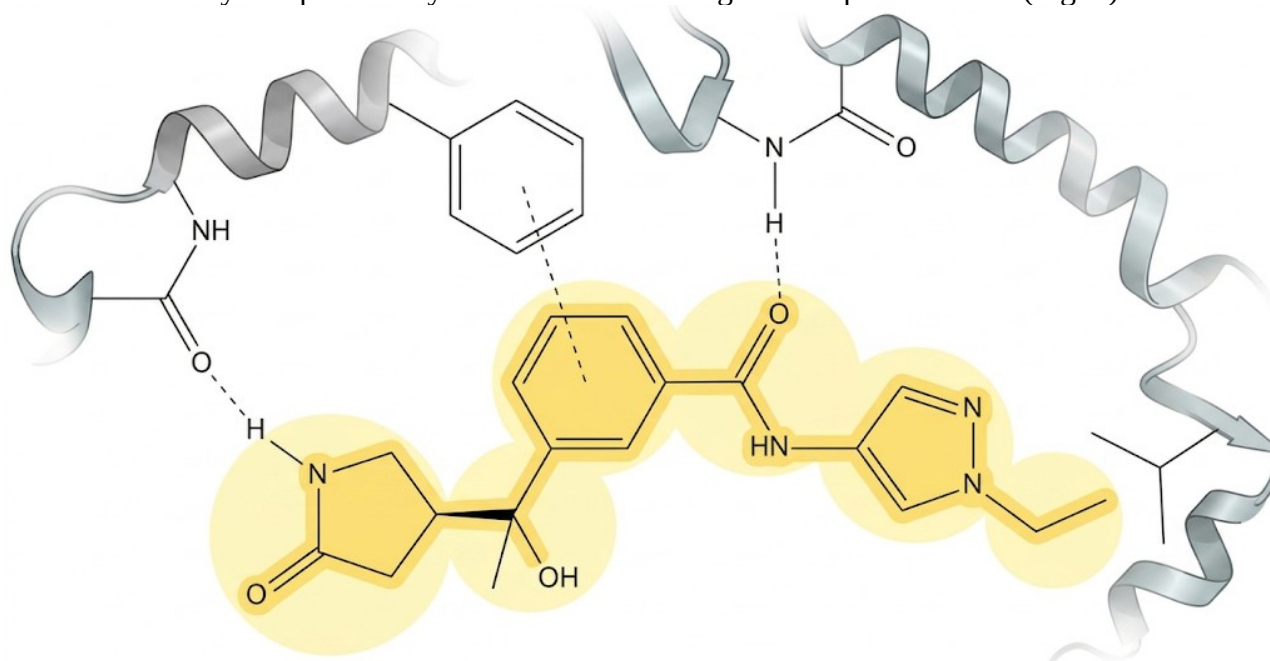


Figure 1

Additionally, the geometry of these interactions is a matter of great importance. Crystal structures of drug-receptor interactions from the Cambridge Structural Database (CSD) and the Protein Data Bank (PDB) showed, for example, that hydrogen bonds follow a specific geometrical rigour such as specific distance, hydrogen bond donor-acceptor angle and a direction of electron lone-pairs. It leads to a precise three-dimensional complementarity to bind and maintain the bonding. If a drug lacks any of these required features, it may form a weak complex with a target or not making it at all. In a structure-based drug discovery, an understanding of these interactions, structural characteristics and conformational capabilities can help with optimising highly potent pharmaceuticals. However, the ongoing challenge is not only how many molecular interactions a drug can make, but also how strong they are in a quantitative sense to establish the overall driving force of binding.¹

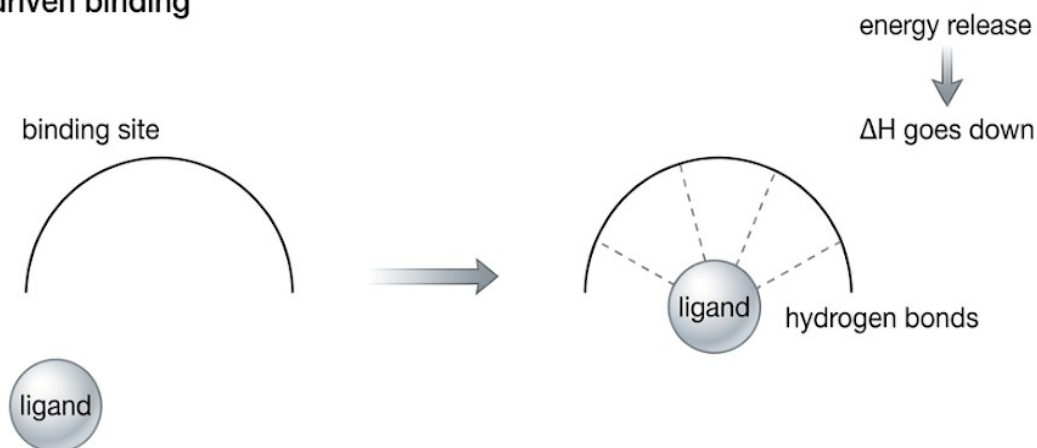
Any spontaneous binding, even the one between a small molecule and its biological target is governed by a negative free energy change (ΔG), which is the sum of change in enthalpy (ΔH), that describes interaction energies (forming and breaking bonds) and the change in entropic component ($-T\Delta S$) that is a measurement of the degrees of freedom of the ligand, protein and surrounding solvent (water) molecules.¹

$$\Delta G = \Delta H - T\Delta S$$

Spontaneous non-covalent binding can only occur when it is associated with a negative binding free energy ($\Delta G < 0$). Free energy components can be complementary or counterproductive, giving a variety of thermodynamic signatures of binding events, ranging from predominantly enthalpy-driven to predominantly entropy-driven.¹

- Enthalpy-driven bond formation with unfavourable entropy has been seen in many complexes with tight hydrogen bonds – which is associated with releasing energy (Fig. 2).¹
- Entropy-driven bond formation, however, with unfavourable enthalpy has been seen purely with hydrophobic interactions – and in this case it is caused (simply putting) by disruption in water molecule organization within the binding site (Fig. 2).¹

Enthalpy-driven binding



Entropy-driven binding

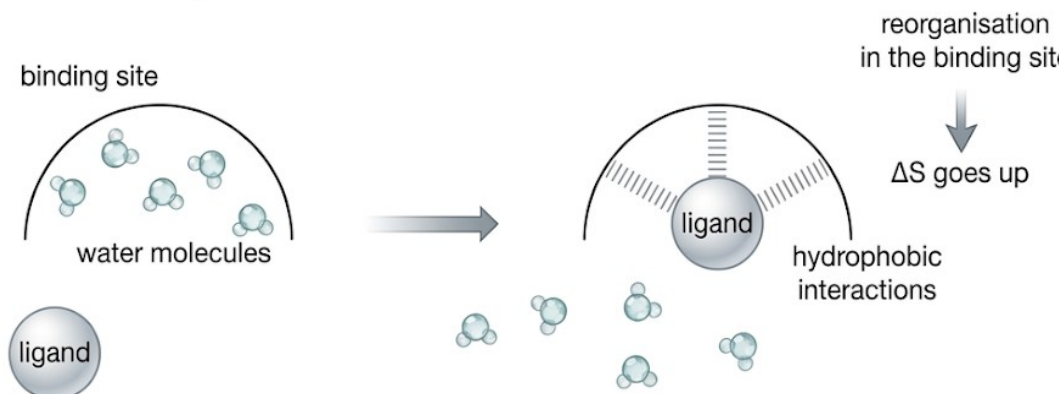


Figure 2. Enthalpy- and entropy-driven binding to a biological target. An intuitive overview.

However, it is rare for drug molecules to rely on either just hydrophobic interactions or only hydrogen bonding, it is usually a palette of various interactions summing together different thermodynamic effects. That is why it is not possible to tell which effect dominates in any given case. Having said that, the best solution in structure-based drug design is optimising ΔG , and not its components individually.¹

1. The most common molecular interactions in drug – ligand complex formation.

Freitas and Schapira analysed over 11 000 X-ray structures of small molecules bounded to proteins from the Protein Data Bank. Their study revealed over 750 000 atom pairs (with a distance

of 0.4 nm or less) that represent molecular interactions and they categorised into seven main groups (Fig. 3).²

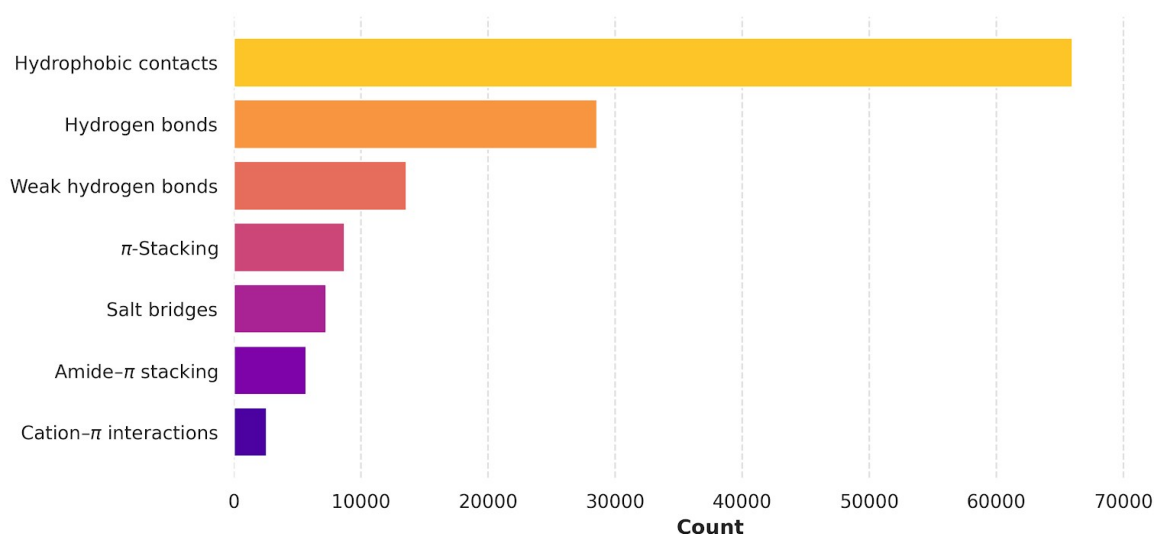
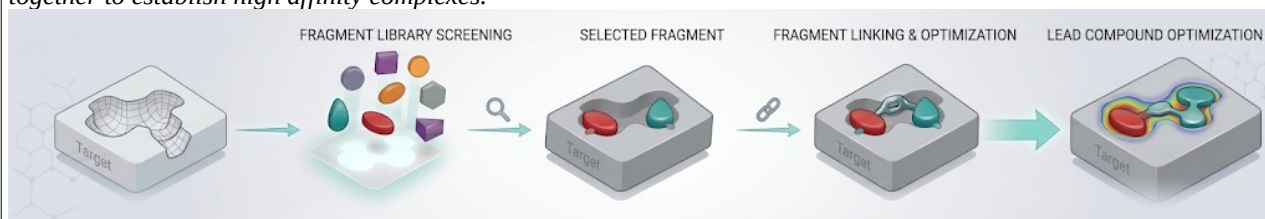


Figure 3. Frequency of non-covalent molecular interactions.

The most common occurring interaction are the well-known types that have long guided drug design such as hydrophobic interactions, hydrogen bonds (including weak ones), π -stacking, salt bridges, amide- π stacking and cation π -interactions. They are complemented by several weak or more specialised interactions less frequently observed in the analysed data set of drug-protein crystal structures, nevertheless, they also play an important role in molecular binding. The massive dataset analysis conducted by Freitas and Schapira raises a fundamental question: why is this the case? Why hydrophobic interactions are predominant in ligand-receptor binding? The authors analysed the relative interaction frequencies of the 1500 protein-ligand complexes with the highest fit quality (ligand efficiency) against the 1500 complexes with the poorest fit quality. The results of this comparison provided the answer: high-efficiency ligands rely more heavily on hydrophobic interactions, while the contribution of others (such as hydrogen bonds and salt bridges) were significantly reduced. Their finding provides a reasonable conclusion that hydrophobic interactions which are modest in strength are structurally and geometrically less demanding than for example hydrogen bonds or electrostatic interactions, thus provide collectively a stronger binding.²

An interesting observation provided by analysis of fragment interactions with protein indicates that much smaller molecules, that are very often used in fragment-based drug discovery (FBDD)* typically rely on high-quality polar interactions. But as the size of ligands grows, lipophilic interactions take the lead. These findings make intuitive sense, as it is much easier for a small molecule to adopt precise geometry to participate in demanding non-lipophilic interactions (fewer degrees of freedom), but at high-level molecular complexity, flexibility favours hydrophobic contacts.²

***Fragment-Based Drug Discovery** – it is a drug discovery approach that identifies small molecules (fragments) as a starting point for drug design. These fragments are usually half or even less the size of typical drugs. Instead of searching in a large collection of drug-like molecules, FBDD uses smaller collections of fragments to establish weak interactions with biological targets, based on which they can identify key structural moieties responsible for signs of low affinity and optimise the structure around them to progressively grow the molecules or linking multiple fragments together to establish high affinity complexes.



Understanding how small molecules bind to their molecular targets is a crucial for the design of new molecules, and it has evolved significantly. Early efforts focus primarily on polar contacts such as hydrogen bonds, electrostatic interactions and others. However, it is clear now that hydrophobic binding is equally important and plays a significant role. Nowadays, the most successful drug discovery strategies try to balance both, stabilising molecular binding through well-placed polar binding, while leveraging hydrophobic complementary interactions to enhance potency and selectivity.²

1.1. Lipophilic interactions

Among all the forces that drive a small molecule-protein binding, lipophilic (hydrophobic) interactions are primary factors in molecular recognition and binding. To put it simply, the mechanism was historically associated with the idea that water molecules occupying a protein pocket are highly organized, similarly to an "iceberg.". When the protein and ligand bind, this "order" is destroyed as water escapes into the bulk solution, creating a massive boost in entropy (disorder) that drives the binding. However, this is an oversimplification. Binding in this case is not just driven by entropy, but enthalpy as well. Within a lipophilic active site, water molecules occupy the space but they cannot form hydrogen bonds with each other properly in an organised way. When they are replaced by a ligand (enthalpy goes up) and water molecules enter a bulk solution, where they can easily form stable network of hydrogen bonds (enthalpy goes down).³ Once the water is displaced and the ligand is nested within the pocket, the close contact between the nonpolar atoms of the ligand and the hydrophobic amino acid side chains (such as leucine, isoleucine, and phenylalanine) triggers close-range van der Waals interactions. Individually, van der Waals forces are relatively weak. However, a lipophilic pocket is structurally complementary to a lipophilic ligand, multiple weak, temporary induced-dipole interactions occur across the entire buried surface area. These cohesive dispersive forces further stabilize the complex.⁴

Aliphatic-aromatic interactions: Lipophilic contacts are dominated by interactions involving carbon, hydrogen or sulphur atoms with a typical distance to biological targets around 4.0Å. The most populated pair of structural moieties engaging in binding are aliphatic carbons (chains) of a target and aromatic rings of a drug. Amino acid side chains in leucine, valine, isoleucine and alanine are the most common participants in these interactions. The contact between them has a preferable geometry, where an aliphatic chain is located above or below of the plane aromatic ring, rather than at its edges (Fig. 5C). Beyond pure hydrocarbon interactions, other important lipophilic types have been identified such as aromatic or aliphatic carbon from the protein and fluorine or chlorine atoms in a drug.

An excellent example of aliphatic-aromatic interactions is inhibition of HMG-CoA reductase, which controls fundamental step of producing cholesterol. Blocking its activity with competitive inhibitors (known as statins) leads to a reduction in cholesterol level and prevents the risk of cardiovascular mortality. Istvan and Deisenhofer obtained a crystal structure of HMG-CoA with atorvastatin and they identify specific hydrophobic interactions. The enzyme undergoes a conformational change upon drug binding, creating a specific hydrophobic pocket. The flat π -electron clouds of atorvastatin's phenyl rings pack tightly *via* dispersive van der Waals forces directly against the branched aliphatic alkyl side chains of Leu562, Val683, Leu853, Ala856, and Leu857. This tight aryl-alkyl alignment prevents the binding of the natural substrate, HMG-CoA.⁵

Aliphatic-aliphatic interactions involve contact between saturated aliphatic chains of a drug and non-polar side chains of amino acids (such as Ala, Val, Leu, Ile). Due to their prevalence in proteins, aliphatic-aliphatic engagements form a massive structural network and are observed in a large number of examples in drug-ligand databases.¹ However, Hans-Jörg Schneide, in his perspective, provides an interesting overview that aliphatic-aliphatic interactions are purely entropy-driven. As simple alkane chains have low polarizability and dispersive (van der Waals) interactions scale with polarizability, aliphatic contact is thermodynamically driven by the cost of returning to water, rather than any effective attraction between aliphatic groups themselves.⁶

Magic methyl: A very common practice in drug discovery is the modification of a compound structure by adding a single methyl group, which can dramatically improve potency.^{2,7a} This approach is often used by medicinal chemists for a strategy known as “methyl walking”, to find a spot in a molecular structure where a methyl plays a crucial role in hydrophobic binding. For example, installation of a methyl group in a tankyrase-2 (TNKS2) inhibitor at meta position to the carbonyl in the aromatic ring, provided almost a 50-fold boost in potency (Fig. 4A).^{7b}

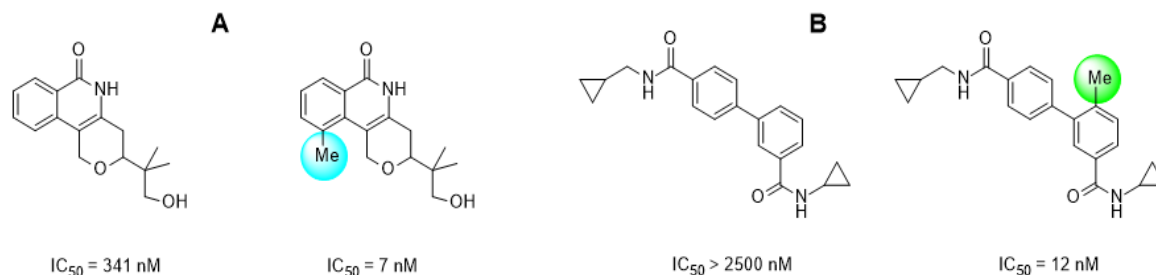


Figure 4. Magic methyl effect in a TNKS2 inhibitor.

Another example is modification of a p38a MAP3 kinase inhibitor that improve potency more than 208-fold. Methylation at para position to the carbonyl led to a low-energy conformer that mimic much better a conformer found in X-ray crystal structure of ligand-inhibitor complex (Fig. 4B).^{7c}

Nevertheless, effect of a single methyl group is very often context-dependent. It may increase overall potency 10-, 100-fold or even more in some cases, but may also have no impact or negatively influence the potency due to steric, conformational, metabolic reasons as well as there is simply no hydrophobic pocket to fill.

Aromatic-aromatic interactions: aryl rings have unique functions in hydrophobic interactions due to their electronic properties. Amino acid side chains of tryptophan, phenylalanine, tyrosine and histidine can interact in aryl-aryl binding with a drug *via* two distinctive geometric forms: T-edge-to-face and face-to-face, which is π - π stacking (Fig. 5). They are highly versatile and unique in drug-target recognition and fundamental for structure-based drug design.¹

T-shaped edge-to-face interactions happen when the edge of one aromatic ring (*via* C-H bond) interacts perpendicularly with the face (π electron cloud) of another ring. It is possible to enhance this interaction by modulating the edge-approaching ring with electron withdrawing groups at ortho or para positions relative to the interacting hydrogen. It can make the charge transfer effect or polarizability-driven inductive effect to strengthen the effect.⁸ T-shaped interaction does not have to be limited to two aromatic rings. Aliphatic protons can also participate with the π electron cloud (the face) and their binding energy can be increased with the acidity of the carbon carrying the proton (e.g. sp -hybridised carbon of alkyne provides much stronger interaction than sp^3 -hybridised alkane carbon or CHF_2 than CH_3).¹

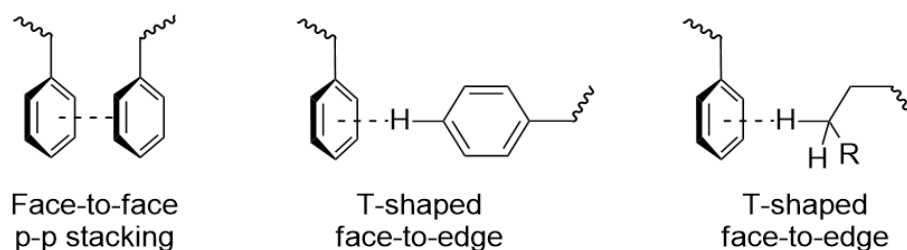


Figure 5. Aryl-to-aryl and aryl-to-alkyl molecular interactions

π - π Stacking (face-to-face) interactions involves two aromatic rings facing each other, stacking parallelly with a slight displacement. This contact is highly dependent on the electronic properties of these rings. Stacking between two electro-deficient rings is generally preferred over stacking of two electron-rich rings. However stacking arrangement between an electron-rich and electron-deficient rings is the most favourable due to profits from charge transfer. Considering heteroaromatic rings, their orientation in stacking interactions is strongly directed by the alignment of molecular dipole and opposing positive and negative charges. Heteroatoms such as nitrogen, oxygen and sulphur reorganise even distribution of electrons. These rings create a parallel-displace stacking and local atomic charge must physically align in complementary fashion (Fig. 6).¹

Beyond traditional aryl-aryl stacking, heteroaryl rings can also be involved in specialised stacking-like arrangements with other functional groups in biological systems like with sulphur. For example a binding site of heteroaromatic adenine rings often interact with the side chain of methionine. $\text{CH}_3\text{-S-CH}_2\text{-}$ unit is frequently found in a coplanar position directly above the purine ring at a van der Waals distance of $4.0\text{\AA}$.¹

In a study of Fa et al., they used molecular dynamics (a computational tool to predict molecules behaviour, movements and interactions) to simulate docking and explored how human Cytochrome P450 inhibitors can bind multiple drug molecules simultaneously. They specifically focused on how face-face stacking interactions support cooperative binding of antifungal drug ketoconazole, that contains five cyclic systems including three aromatic rings. Researchers observed that when one molecule binds to the enzyme Phe215 forms a π - π stacking with piperazine ring of the drug. Adding an additional ketoconazole molecule, Phe215 acts as an anchor, forming an aryl-piperazine-piperazine sandwich system supported by Phe304 (Fig. 7).^{9a}

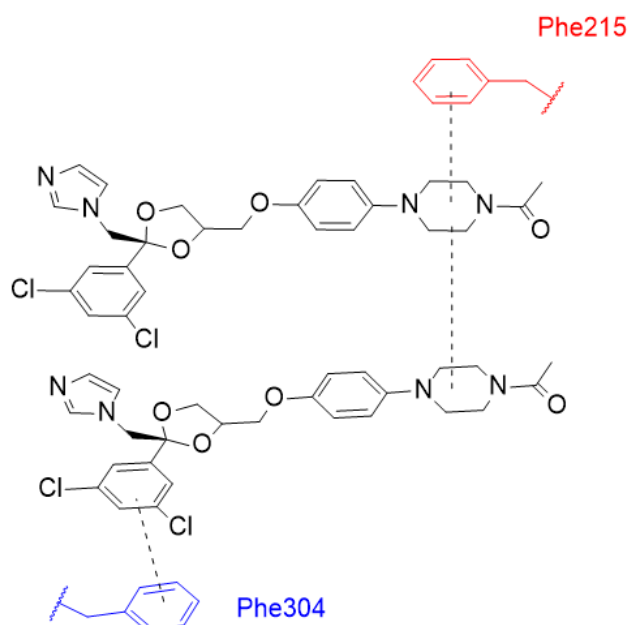
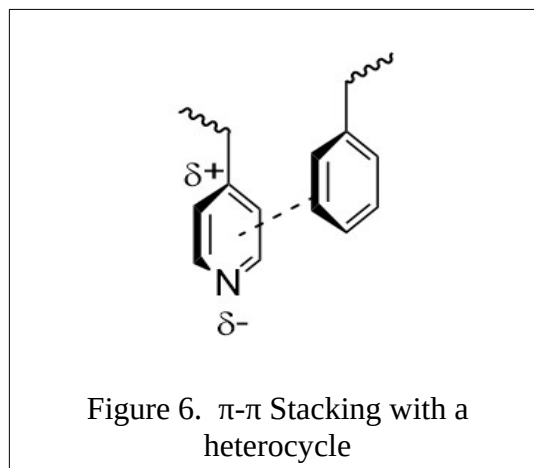
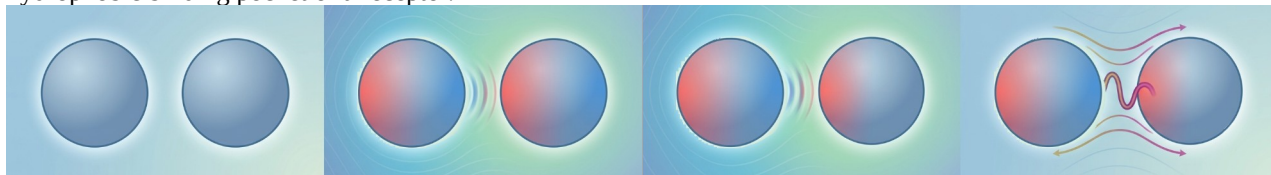


Figure 7. A schematic representation of multiple π - π stacking of two ketoconazole molecules with CYP3A4w in molecular dynamic docking.

Van der Waals force are weak, short-range attractive forces that occur between electrically neutral molecules. This interaction are related to three main phenomena such as interactions between permanent dipoles, a dipole and induced dipole and the most importantly London dispersion forces. The latter one are transient shifts in electron density to create short-lived complementary dipoles in adjacent molecules, generating attractions between them. Even though they are weak individually, as a collective become substantial across large contacting surfaces found for example in drug-receptor interactions. That is why they are the key importance in lipophilic binding where a ligand is nested in a hydrophobic binding pocket of a receptor.¹⁰



In a high-throughout screening Eldrup et al. identified a promising soluble epoxide hydrolase (sEH) inhibitor.^{9b} However, due to short metabolic half-life and rapid clearance, the compound required hit-to-lead optimisation. Researchers hypothesized that removing one phenyl group might be beneficial for physicochemical properties such as solubility (Fig. 8). However, structure-activity relationship revealed a great importance of diphenyl fragment. A monophenyl derivative was 100-fold less potent than the original hit. X-ray cocrystal structure of sEH with the compound showed additional π - π stacking with His524 supporting similar interaction with Trp 336 and pyridine ring.

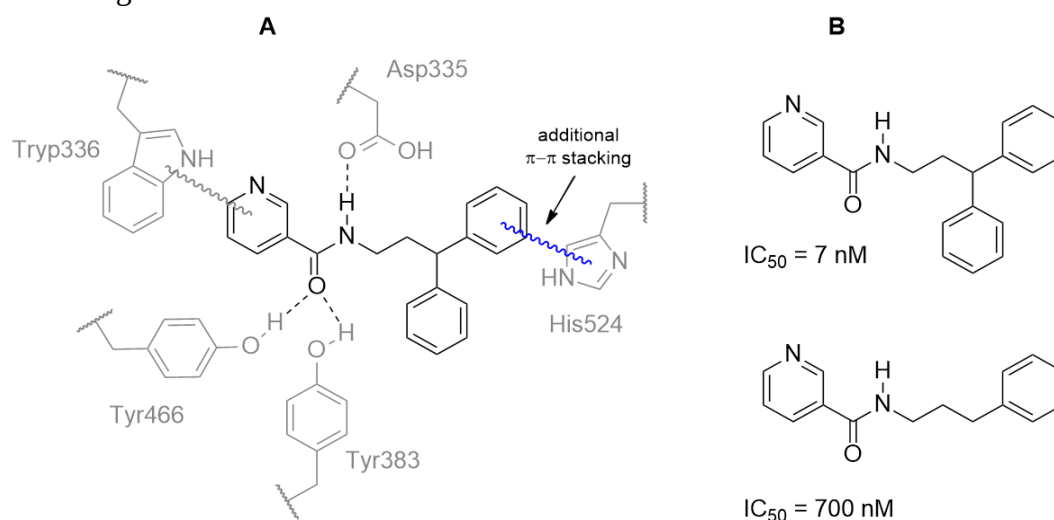


Figure 8.

1.2. Hydrogen bonds

Hydrogen bonding operates at the threshold between weak non-covalent interactions and robust covalent linkages, ranging from 1 to even 40 kcal/mol (for example in $[F \cdots H \cdots F]$). With its variations in strength and defined directionality, it is the foundation in molecular interactions and forming complex architectures. Over the last century, hydrogen bonding has transformed from an elusive concept to a well-defined molecular binding. It started back in 1920 when Latimer and Rodebush inspired by Lewis's valence theory, proposed that a hydrogen nucleus can be held by two electronic octets forming a weak tricentric structure. Later on Pauling proposed an electrostatic, dipole-dipole explanation in 1930s. In 20th-century the consensus that hydrogen bonding dictates three dimensional architecture, accelerated discoveries in revealing the helical structure of DNA, α -helix and β -sheet structures of proteins and organised patterns in organic crystals. Nowadays, hydrogen bond formation finds applications in the areas of modern science such as molecular

recognition, catalysis, assembly that have been intensively explored by organic and supramolecular chemists.^{11a}

A classical hydrogen bond (X-H...Y) can be described two ways: as a covalent-like donor-acceptor orbital interactions ($LP_Y \rightarrow \sigma^*_{X-H}$) or an electrostatic dipole attraction ($X^{\delta-}H^{\delta+}\cdots Y^{\delta-}$). This explains why a O-H...N bond is stronger than N-H...N bond: the higher the electronegativity of the heteroatom carrying the hydrogen, the stronger the polarization of the X-H bond, which increases both the positive partial charge on the hydrogen atom and the efficiency of the orbital interaction.^{11a}

The strength and geometry parameters of hydrogen bonds can vary significantly. A typical hydrogen bonding (N-H...O or O-H...O) usually follows steric distance requirement of 2.7 to 3.1 Å. Yet, hydrogen bonds represent a broad energetic spectrum from weak (barely 1 kcal/mol) to exceptionally strong like in low-barrier hydrogen bonds (LBHBs)¹² where the proton is shared between a donor and acceptor equally (Table 1). Furthermore, the concept of hydrogen bonding has expanded to less conventional interactions such as C-H...O or π -mediated (C-H... π and O-H... π), which provides a further diversity in the ways that molecules can assemble in space. Due to steric geometry requirements and directional character, hydrogen bonds are a matter of great importance in biological systems and because of that they are widely used in ligand-receptor binding studies and structure-based drug design.¹

It is important to point out that, hydrogen bonds offer specific interactions rather than a robust linkage providing a massive gain in overall binding free energy. Due to the fact that ligand-receptor binding occurs in an aqueous environment, it is more a hydrogen bond exchange rather than formation. Before a ligand can bind to its biological target, both of them must break pre-existing interactions with surrounding water molecules that stabilise their structure. Consequently, hydrogen bond formation and dehydration largely offset one another due to enthalpy-entropy compensation, which is an intrinsic feature of binding in aqueous solution. The net gain in binding free energy may be negligible or even unfavorable unless the newly formed polar interactions are of sufficient strength and geometric complementarity to overcome this compensation barrier. For that reason, the main task for medicinal chemists is not to maximize the total number of hydrogen bonds, but to ensure that added and existing donors and acceptors within an active site are geometrically corresponded and both HBDs and HBAs provide as strong hydrogen bonding as possible. Keeping unnecessary polar groups in a dehydrated binding pocket without a complementary partner may lead to decrease in the affinity of a ligand towards the receptor.^{1,11b,c}

Table. 1 Strength, Length and angle of chosen hydrogen bond types.

Description	Very strong	Strong	Weak
Length [Å]	2.2-2.5	2.5-3.2	3.0-4.0
Bond energy [kcal/mol]	15-40	4-15	>4
X-H...Y angle	175-180	130-180	90-180
Examples	[F...H...F] ⁻ [N...H...N] ⁺ P-OH...O=P	O-H...O=C N-H...O=C O-H...O-H	C-H...O N-H...F-C O-H... π

During optimisation of a drug's structure, chemists must carefully balance the roles of hydrogen bond donors (HBDs) and acceptors (HBAs). Even though traditional guidelines like Rule of Five suggest limits of HBDs (5) and HBAs (10), detailed physicochemical analysis revealed that HBDs are generally less polar and less hydrated than HBAs (Figure 9).^{13a,b} This means that desolvation penalty alone does not explain why HBDs can be considered problematic in drug design, as their number is half that of HBAs. The true problem with utilising HBDs in molecular recognition lies in their dual character, they are HBDs but they also may play a role of HBAs as well (Fig. 10)¹⁴

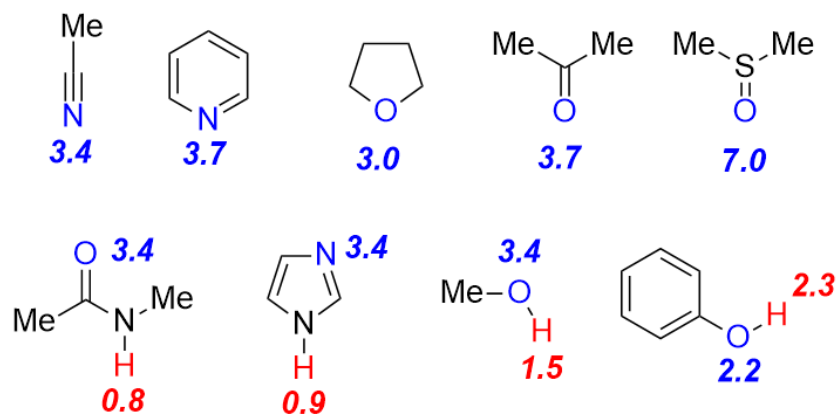


Figure 9. Calculated values of polarity (LogP_{alk_cal}, based on hexane/cyclohexane-water partitioning). An example: polarity (q) of carbonyl in amide was calculated as follow: $q(\text{C=O}) = q(\text{N-methyl acetamide}) - q(\text{N-methylethylamine})$.

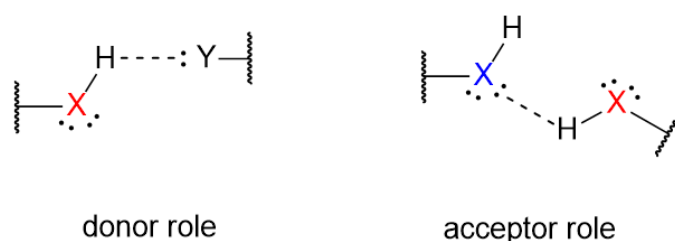


Figure 10. Dual character of hydrogen bond donor (HBD)

While an oxygen-based donor like a hydroxyl group introduces a small steric footprint, but it brings along an oxygen atom that concurrently acts as a hydrogen bond acceptor. If the target receptor's pocket lacks a corresponding donor to satisfy this oxygen, the resulting desolvation penalty can entirely negate the affinity gained by the initial hydrogen bond. To avoid this, medicinal chemists frequently utilize nitrogen-based donors, such as aromatic NH groups found in pyrazoles or indoles. These groups offer "line-of-sight" access to target HBAs without presenting an unsatisfied acceptor under normal physiological conditions, making them highly efficient elements for target recognition. Additionally, when multiple hydrogen bonds are properly aligned at the ligand-receptor interface, they can benefit from frustrated hydration and secondary interactions that may enhance overall binding affinity. However, replacement of hydroxy group with for example pyrazole may influence lipophilicity and also bring a risk for CYP or hERG inhibition.¹³ That is why a careful consideration and study overall physicochemical and pharmacological properties are needed.

An impressive example of structure modification by adding a hydrogen bond donor (amino group that in aqueous condition forms an ammonium salt) is a thrombin inhibitor. Replacing 3-phenylcarboxylate with phenylalanine led to formation of an additional hydrogen bond with Gly216 (Fig. 11) that led to remarkable increase in potency (>500-fold).^{15a,b}

Thrombin is an enzyme responsible for transforming fibrinogen into fibrin and activation of blood platelets that lead to blood clotting. Thrombin inhibitors are a group of anticoagulant medications that selectively block thrombin activity in clotting. They are prescribed for the prevention and treatment of venous thromboembolism, prevention of strokes and the treatment of acute coronary syndromes

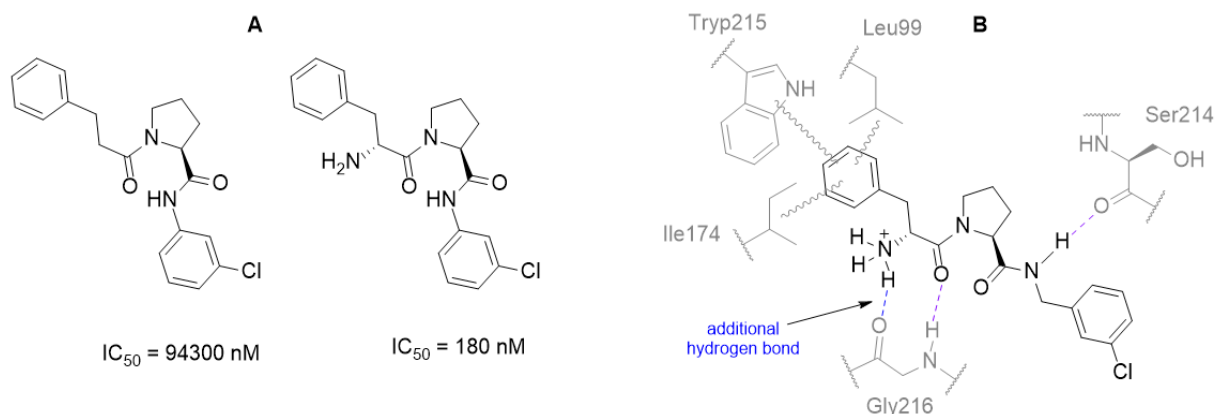


Figure 11. A) Boost in potency by introducing an amino group; B) schematic representation of the aminoinhibitor binding to thrombin enzyme.

Later on other Nasief, Said and Hangauer investigated the SAR to determine the effect of hydrogen acceptor strength on the binding affinity to thrombin.¹⁶ It seems to be an intuitive expectation in drug design that the stronger the hydrogen bond acceptor, the stronger the resulting hydrogen bond with the biological target that would result in a better binding affinity (Fig. 12), and this is an absolutely legitimate assumption. However, the researchers demonstrated that this systematic reasoning was incorrect for the class of thrombin inhibitors investigated.

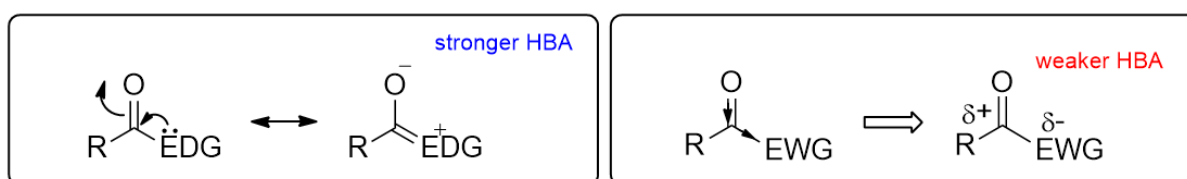


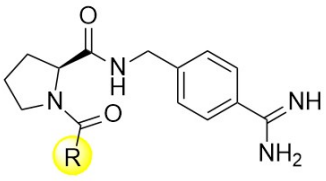
Figure 12. The influence of electron withdrawing (EWG) and electron donating (EDG) groups on the carbonyl as a hydrogen bond acceptor (HBA).

They obtained a series of analogues containing various substituents at the carbonyl which forms a hydrogen bond with Gly216 of the enzyme and observed the reversed form expected influence of R on the binding affinity to thrombin (Table 2). The electron donating groups (R = NHBU^t , NH_2) that were supposed to increase capability of carbonyl to form stronger hydrogen bonds and enhance binding affinity, had a negligible effect. Electron withdrawing group (R = SO_2CF_3 , CF_3), on the other hand, dramatically improved it. Based on their mechanistic explanation of this phenomena the reason is the desolvation penalty. A more basic hydrogen bond acceptor (R = NH_2) binds water molecules more tightly in solution. When a ligand binds to the protein, those water molecules must first be stripped away (the desolvation penalty). The cost is much lower for less basic hydrogen bond acceptors (like SO_2CF_3 , CF_3). The authors made a case that for exposed hydrogen bond acceptors at the binding site of a protein, desolvation penalty is not a subtle effect, but in some situations it can be a predominant result on the overall binding affinity.

$$\Delta G = \Delta G(\text{protein-ligand interaction}) - \Delta G(\text{desolvation}) + \text{others}$$

This work is a remarkable example that medicinal chemists need to look for explanation during structure-activity relationship analysis, especially if the data is unclear or it gives misleading results.

Table 2.

					
A			B		
R =	ΔG [kJ/mol]	Ki [μM]	R =	ΔG [kJ/mol]	Ki [μM]
NHBu ^t	-25.4	36	CH ₃	-23.8	68
NH ₂	-24.1	59	CH ₂ F	-24.9	44
SO ₂ NH ₂	-21.6	163	CHF ₂	-27.1	18
SO ₂ Me	-23.8	68	CF ₃	-27.8	13
Me	-23.8	68	-	-	-
SO ₂ CF ₃	-29.4	7	-	-	-

An article by Lawhorn et al. represents a medicinal chemistry study of electronic modulations that can tune drug-receptor hydrogen bonds to troponin I-interacting kinase (TNNI3K) as the primary target.¹⁷ They studied a series of substituted N-methyl-3-(pyrimidin-4-ylamino)benzenesulfonamide inhibitors that by X-ray crystallography show hydrogen bonds to the kinase domain *via*: a) the nitrogen of the heterocyclic ring that acts as HBA to the backbone of Ile542 and b) the sulphonamide that acts as HBD to the side chain of Thr539. The strongest hydrogen bonding occurs when Y is a powerful electron-donating group such as amine or methylamine. On the other hand, sulphonamide group becomes more acidic (a better hydrogen bond donor) if X is an electron withdrawing group such as Cl or OCF₃ (Fig. 13).

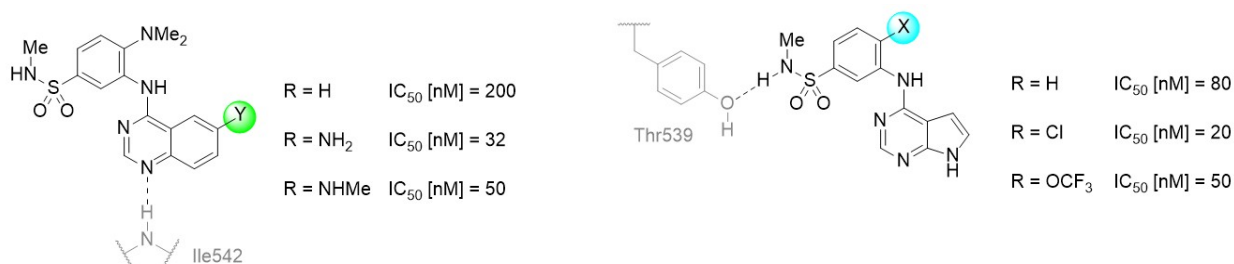


Figure 13. The influence of EWG and EDG groups on hydrogen bonding and potency.

Summarizing, hydrogen bonding is an important factor in drug-receptor binding providing specific interactions for molecular recognition. During structure-based drug design it is essential to recognise possible hydrogen binding sites (for example *via* computational docking – CADD) and modify the drug's structure effectively to optimise proper distance and angle (conformational factor) and their strength (electronic factor) for high affinity to a biological target.

1.3. Weak hydrogen bonds

Unlike typical hydrogen bonding, weak or non-classical hydrogen bonds involve less electronegative donors such as C-H groups, or unconventional hydrogen bond acceptors such as aromatic π systems or halogens. Due to their weak character, they exhibit longer interaction distances and broader geometric factors. For example, the median distance of C-H $\cdots$ O bond is

approximately 3.4Å, which is roughly by 0.4Å longer. C-H...O bond is the most populated weak hydrogen bond observed in major database analysis of protein-ligand complexes. Strength of this bond is highly dependent on the acidity of the donor, which can be increased by appropriate substitution, for example fluorine atoms at C $^{\alpha}$ of aliphatic donors or EWG at ortho or para positions in aromatic donors. The same situation can be observed in non-classical HBAs. By increasing electron density of aromatic rings, they form much stronger linkage with polar hydrogen bond donors (such as N-H or O-H) (Fig. 14).¹

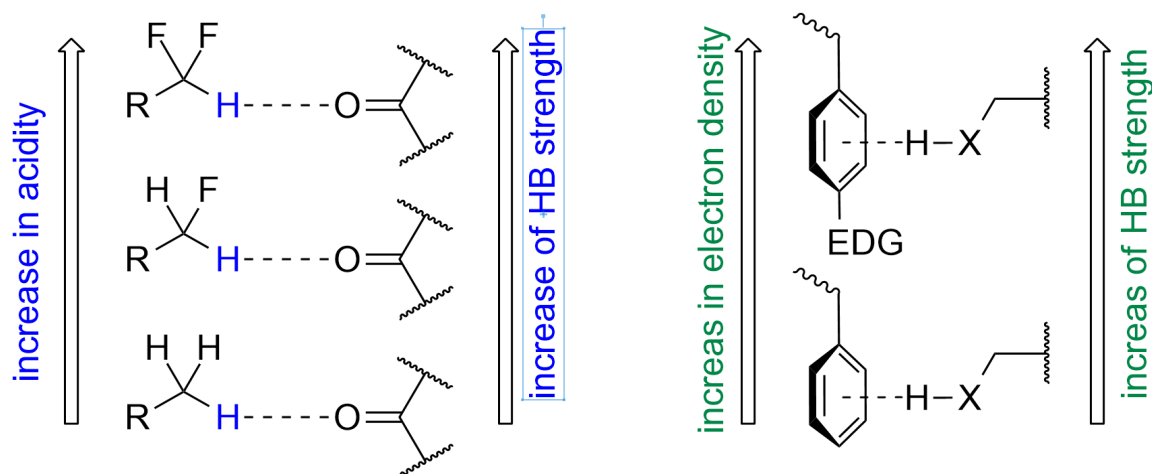


Figure 14. The influence of substitution on the strength of non-classical HBDs and HBAs.

A huge advantage of these interactions in drug design is their highly favourable desolvation profile. As it was discussed previously, typical polar groups require stripping away tightly bounded water molecules before they can engage in binding with the biological target. This causes a thermodynamic penalty that often cancels out the energy gained from the newly formed hydrogen bonds. In contrast, weak HBDs and HBAs are barely hydrated, which means their desolvation cost is minimal. Having said that, medicinal chemists can sometimes increase drug's overall affinity binding by replacing classical polar groups with their weak equivalents, because the energy saved by decreasing desolvation penalty can bring an overall gain.

A notable example of the advantages of weak hydrogen bonding over a classical one is CDK2 inhibitor (Fig. 15). A matched pair illustrates that even though amino group forms a stronger hydrogen bonds, the desolvation penalty is a crucial factor that influence the binding affinity. Thus, introducing the methyl derivative to enable weak hydrogen bonding ultimately yielded a more potent inhibitor.¹⁸

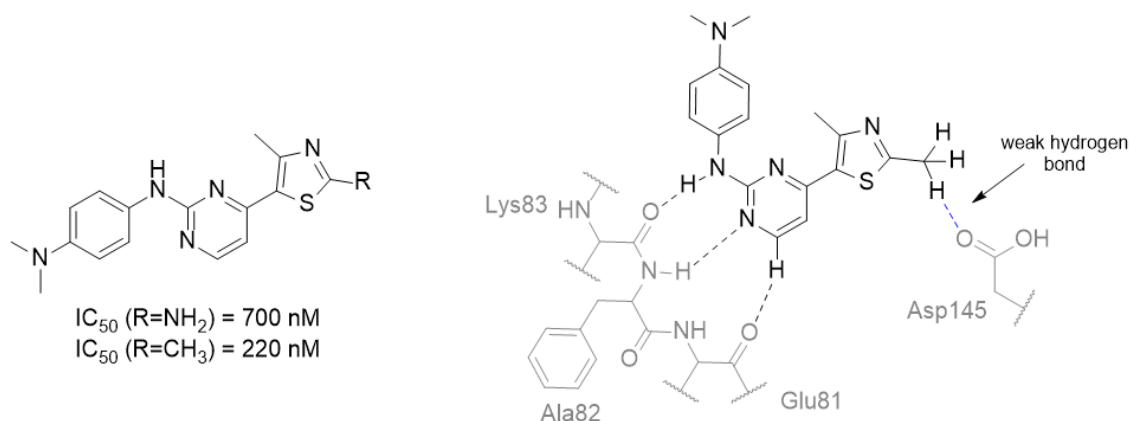


Figure 15. Potency of two CDK2 inhibitors and a representation of molecular interactions for R=Me derivative.

Panigrahi and Desiraju investigated a library of amidinium-based inhibitors of Human African Trypanosoma (parasites) which are responsible for Human African Trypanosomiasis (sleeping sickness).¹⁹ In addition to van der Waals forces, hydrophobic effect, strong hydrogen bonds and their weak version are assisting interactions that support the binding geometry to the DNA of these single-cell organisms, causing cytotoxic effect. For example C-H groups form weak hydrogen bonds with nitrogen atoms of purines N(3), pyrimidines O(2) and deoxyribose sugar moiety of the DNA part O(4') (Fig. 16).

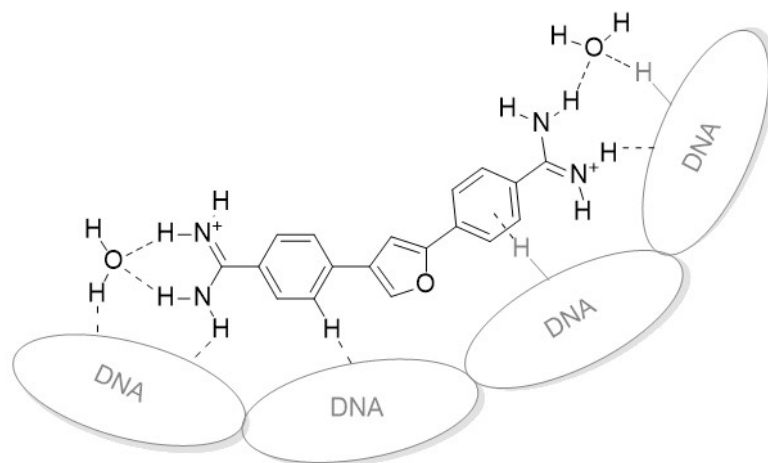


Figure 16. A schematic representation of an amidinium-based inhibitor bound to a DNA chain of Human African Trypanosomes.

Burton and co-workers provided the synthesis and biological evaluation of new steroid analogues to target DAF-12 receptor in nematodes. Derivatives of the steroidal hormones dafachronic acids might have potential application in curing parasitic worm infections.²⁰ They identified a lead compound (a primary alcohol) that has a residual agonistic activity. During SAR study they replaced hydroxyl group with CHF₂ to evaluate its binding properties hoping that this bioisostere would maintain a hydrogen bond donor, and provide better drug-like properties. As the study showed, difluoromethyl analogue preserved similar activity (maintained HBD property) but authors suspect that removing highly hydrophilic OH might have provided improved absorption and extension in half-life (Fig. 17).

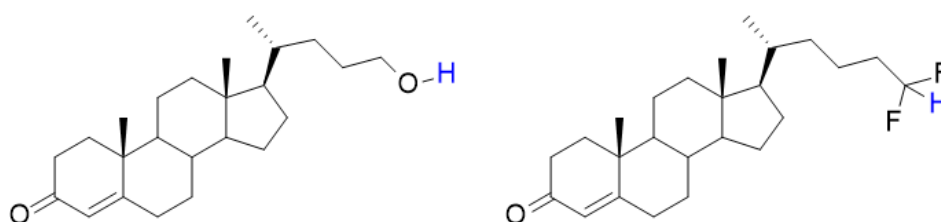


Figure 17. Bioisosteric replacement of OH group by CHF₂ to maintain a hydrogen bond donor.

Parasitic worms infect billions, but existing drugs rely on outdated mechanisms that fail to target the critical infection stage. Nuclear receptors are ligand-gated transcription factors regulating metabolism and development. They are proven mammalian drug targets also being explored in parasites. Specifically, the DAF-12 receptor, first identified in *C. elegans*, has emerged as a promising new target for anthelmintic drug discovery.²¹

1.4. Salt bridges

Salt bridges are ionic molecular interactions that occur between positively charged atoms (e.g. nitrogen in a ligand) and negatively charged ones (such as a carboxylate oxygen in a protein). These ionic linkages act as a charged-assisted hydrogen bonds and are one of the most populated interactions in protein-ligand complexes.^{1,2} Their short interaction distances may indicate highly strong bonds, however, incorporating salt bridges does not automatically guarantee a favourable boost in binding affinity. Charged functional groups are heavily hydrated in aqueous environments with a dense network of hydrogen bonds with water molecules. Forming a salt bridge connection with a protein at its active site requires a massive desolvation penalty. In solvent-exposed areas on the protein surface, the energy gained by pairing two charged ions merely offers the thermodynamic cost due to stripping away surrounded water molecules, offering little or no gain in binding free energy. Therefore, to successfully exploit salt bridges in structure-based drug design, the interactions need to be partially buried within the protein pocket and stabilised by other surrounding interactions. Under these conditions, salt bridges can even become a crucial part of high affinity binding between a receptor and its ligand.^{1,2}

Based on the Drug Bank the most populated anionic partners of salt bridges in receptor-ligand interactions are carboxylates, monoesters of phosphates, and phosphonates. The most frequently occurring cationic species in this interaction are primary, secondary and tertiary ammonium salts (Fig. 18).²²

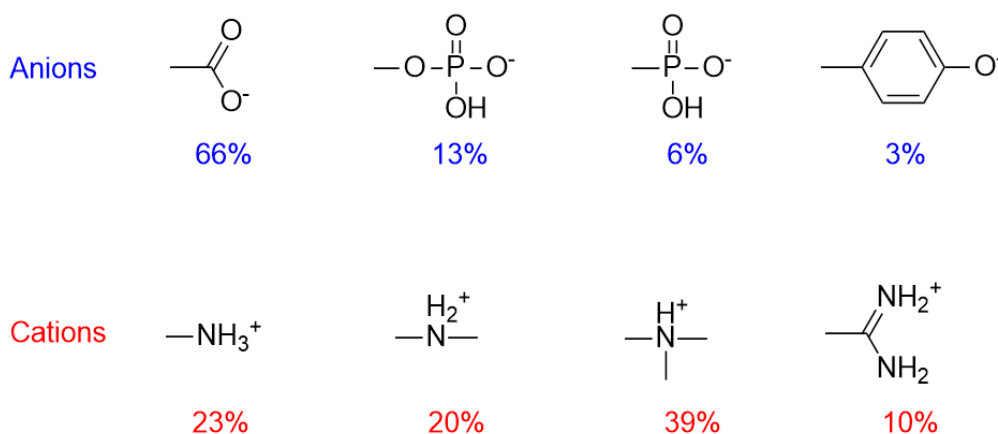


Figure 18. The most populated cations and anions participating in salt bridges based on the Drug bank.²²

The strength of ionic bridges strictly depend on the binding environment and they may have an enormous influence on ligand-receptor interaction affinity. An example of such binding is an ammonium derivative that forms a supported ionic linkage with a side chain of aspartic acid (Asp831) in the kinase domain of epidermal growth factor receptor (EGFR), resulted in an enormous 800-fold boost in binding affinity (Fig19).²³

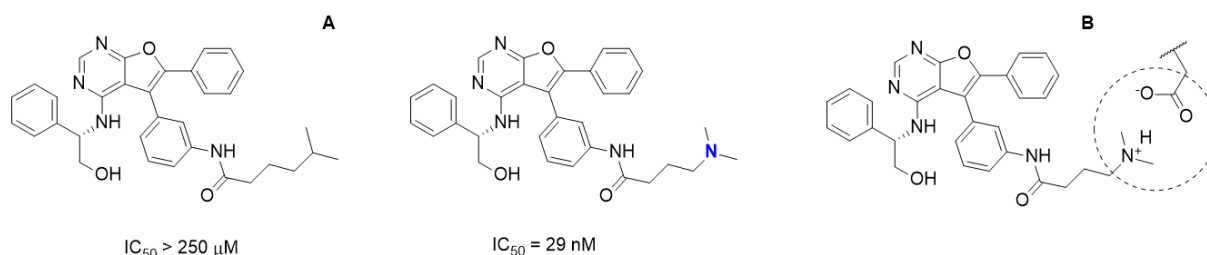


Figure 19. A) Difference in binding affinity to EGFR; B) A schematic representation of ionic interaction

1.5. Amide π -stacking

This weak interaction represents an interesting but still underused strategy in structure-based drug design, ranking as the 6th most frequently observed non-covalent binding in biological complexes. It occurs when the π -electron surface of an amide bond stacks against the π -surface of an aromatic ring. The interaction exploits the dual electronic nature of amide bond. Amides with their rigid structure often participate in hydrogen bonding, they also remain polar and highly polarizable in the perpendicular direction, making their π electrons excellent stacking partners (Fig. 20).^{1,2}

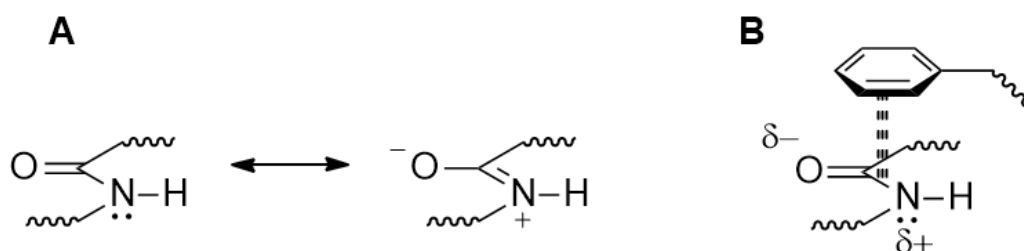


Figure 20. A) Dual nature of amide group; B) face-to-face amide π stacking interaction.

In terms of geometry, this interaction is quite versatile. It occurs with similar probability in both face-to-face and edge-to-face arrangements with a typical distance between 3.2 to 3.7 Å. There are distinct amino acid preferences depending on the geometry. Glycine equally forms both geometric arrangements, while tryptophan prefers face-to-face and leucine edge-to-face. From a medicinal point of view, the most striking feature of this interaction is that over 88% of them take place between side chains of glycine and aromatic rings of the ligands. As amino acids are the fundamental building blocks of protein, identifying an exposed amide group in a binding pocket provides a design opportunity, an additional anchor point to position an aromatic ring and proactively increase target affinity.²

An unexpected discovery was made by Salonen and co-workers during development of new factor Xa inhibitors as potential antithrombotic agents to treat or prevent blood clots.²⁴ They observed that properly oriented oxazole ring in the structure of inhibitors was the key for enhancing binding affinity. The authors discovered that the interaction's strength is heavily influenced by dipole alignment of the heterocyclic ring and amide bond. A simple trick by flipping the orientation of the oxazole ring to align its molecular dipole antiparallel to the amide's dipole resulted in 11-fold increase in potency (Fig. 21).

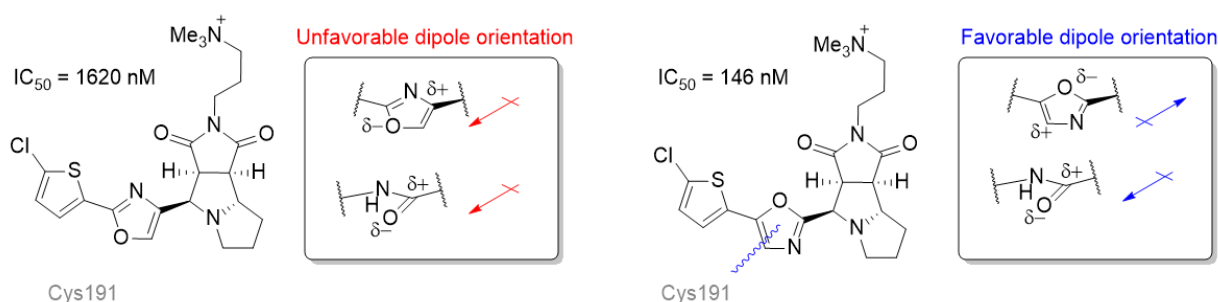


Figure 21. The influence of oxazole orientation on potency.

1.6. Cation- π interactions

As previously described, the electron-rich aromatic rings possess a partial negative charge, allowing them to engage in electrostatic interactions with partially positively charged structural fragments such as protons and amide bonds. This is also possible with formally charged cations *via* π -cation interactions. It is extensively observed especially in protein structure stabilization but also in target-ligand recognition. This engagement can occur in both ways: positively charged side chains of amino acids with aryl rings of a ligand as well as positively charged ligand with aromatic rings of amino acid side chains (Fig 22).^{1,2}

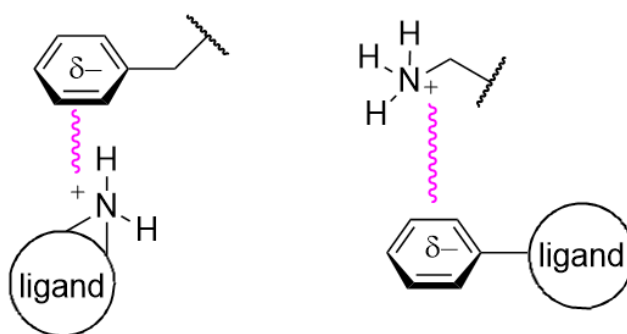


Figure 22. π -cation interactions.

When the protein provides the cation, arginine side chains are observed to form this interactions about 3 times more often than lysine. This preference comes from arginine's guanidinium group that can simultaneously participate in other type of interactions (such as hydrogen bonding). When protein provides aromatic π -system, it is typically tryptophan and tyrosine.

Medicinal chemists can use this interactions to modulate target affinity by structure modification and minimize the desolvation penalty. A great example is the optimization of previously described factor Xa inhibitors but at a different part of the molecules.²⁵ Methylation of amine group preserves positive charge at the nitrogen atom, but the more methyl groups are introduced the lower hydration penalty is observed, which has a major impact on the binding affinity to the factor Xa protein (Figure 23).

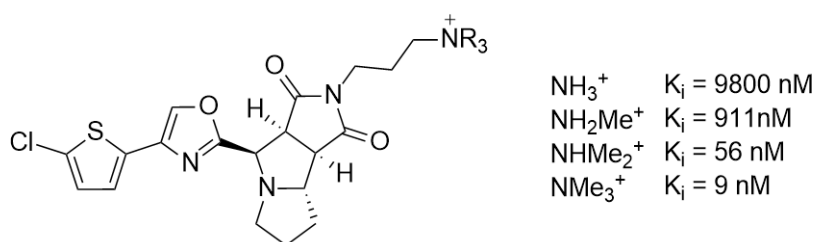


Figure 23.

In cation- π interaction a fundamental factor is a presence of a positive charge in one of the engaging molecules. Without the charge, naturally the binding cannot occur. A perfect example is a mechanistic study and key interactions of varenicline and its analogues with the nicotinic acetylcholine receptor.²⁶ One of the most important features of varenicline is its protonation state, because the secondary amine takes part in both hydrogen bonding and cation- π interaction with TrpB. Experimentally determined pK_a of varenicline is 8.90, but its pyrazino[2,3-b]pyrazine derivative only 7.31. The authors suggested that the tetraaza derivative's lack of activity comes from the low pK_a of the piperidine amine, which means the amine is poorly protonated at physiological pH. Protonation of the piperidine nitrogen is crucial for the cation- π and the H-bond donor interactions. If the amine isn't protonated, those interactions are severely impaired (Fig. 24).

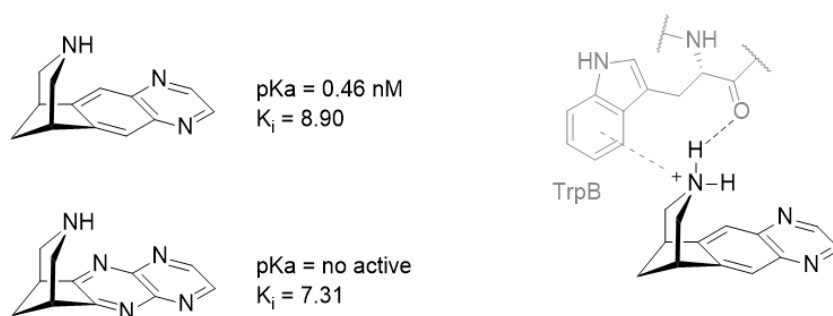
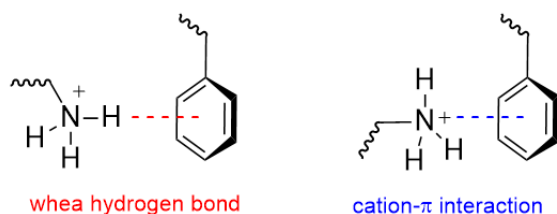


Figure 24. Varenicline and its tetraaza derivative.

The difference between a weak hydrogen bond between positively charged amine and aromatic ring AND a cation- π interaction between positively charged amine and aromatic ring. The amino group acts as a polarized X-H structure (H-donor). The hydrogen atom points directly toward the face of the electron-rich π -system (H-acceptor). It is driven mainly by electrostatic energy and is classified as a weak hydrogen bond with lower stabilization energies ranging between 6 and 7 kJ mol⁻¹. While, in cation π interaction, instead of interacting via the intermediate hydrogen atom, the interaction happens as a direct engagement between the aromatic ring and the actual cationic center (N⁺) of the ammonium molecule. This interaction is significantly stronger and can reach between 40 and 75 kJ mol⁻¹.



2. Other molecular interactions

Beyond the classical hydrophobic interactions, hydrogen bonding, salt bridges and hydrophobic contacts there is a broader group of less common or “unconventional” noncovalent interactions, there are others that still can be found or applied in structure-based drug design. Even though they might be weaker and require more specific structural, special or electronic conditions, they play important roles in protein folding, enzyme catalysis and are available for medicinal chemists to enhance drug-target molecular recognition.

2.1. σ -hole interactions

It is a class of interactions that are based on σ -hole, which is an electron-deficient positive charged region of a strongly polarised heteroatom that is located on the opposite side to a covalent bond (Fig. 25). There are a few types of this binding:^{1,2}

- Halogen bond, which is formed between halogens (chlorine, bromine, iodine) and electron-rich nucleophiles like oxygen or sulphur. Fluorine rarely participates in this type of bonding due to its high electronegativity and lack of polarizability. In drug design, halogens can be used to mimic hydroxyl groups, fitting into narrow binding pockets without desolvation penalty of an oxygen-containing acceptor.
- Chalcogen bond is formed by group VI elements, primarily sulphur and selenium, which interact via their own σ -holes. They have an electrostatic origin where the charge transfer occurs from the donor (nucleophile) to the antibonding orbital of chalcogen atom (σ^*). For example,

sulphur in methionine residue can form highly directional attractive contacts with the π region of carbonyl oxygen, stabilizing protein structure.^{1,2,27}

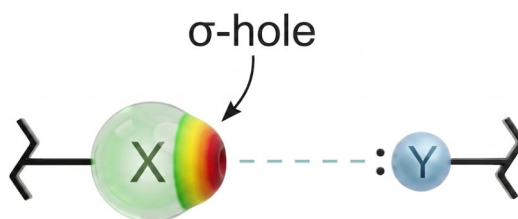


Figure 25. A graphic representation of σ -hole interaction.

2.2 Anion-aryl interactions

While cations can interact with electron-rich aromatic rings, negatively charged ions or groups can also be involved in interactions but with electron-deficient aromatic π -systems. This type of linkage may happen via anion- π interactions, where a negatively charged ion is attracted by an electron-deficient region of an aromatic ring at the face site. Far more common in proteins is the interaction between anion and the edge (C-H bonds) of an aromatic ring (anion quadrupole interaction). The edge-on interaction avoids the unfavorable electron-rich p cloud and driven by electrostatics, this interaction can reach up to -8.0 kcal/mol (Fig. 26).²⁷

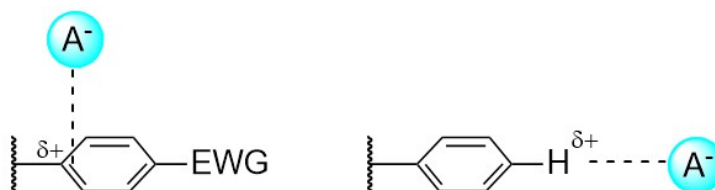


Figure 26. anion- π and anion-quadrupole interaction.

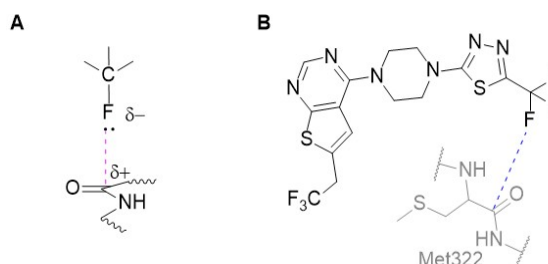
2.3 Carbonyl-carbonyl interactions

This type of bonding is often seen in proteins. When the lone pair (n) of one carbonyl oxygen delocalises into the antiorbital (π^*) of a neighbouring carbonyl group. These arrangements frequently occur within α -helices to help stabilize protein secondary structures.²⁷

2.4. Orthogonal multipolar interactions

It was found between two dipolar functional groups approaching each other in a perpendicular way. A prominent example is the orthogonal interaction of a fluorine substituent (C-F) to the carbonyl. This geometry neutralises dipole-dipole repulsion, allowing higher-order electrostatic dispersion forces to create a weak attractive interaction that has been successfully utilized to boost drug affinity (Fig. 27).^{27,28}

Figure 27. A) A general overview of orthogonal multipolar interaction; B) A graphic representation of fluorine orthogonal multipolar interaction of a ligand with MI-2-3.²⁸



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